

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and executed complex experiments for reagent systems and assay development, integrating customer workflows and regulatory requirements.
- Analyzed experimental data using **statistical methods** and presented findings to cross-functional technical teams and scientific review boards, influencing product development decisions.
- Investigated, troubleshooted, and resolved technical issues in laboratory procedures, ensuring robust product development and maintaining **99.9% experimental uptime**.
- Developed and maintained clear, accurate **protocols**, technical reports, and study summaries in compliance with **good laboratory practice (GLP)** standards.
- Collaborated with cross-functional teams, including R&D, Quality Assurance, and Manufacturing, to achieve program milestones and product launch commitments, mentoring junior team members on experimental techniques.
- Applied **scientific judgment** and critical thinking to advance product development for analytical instruments and workflows, contributing as a core team technical expert.

SKILLS

Scientific Domain – Immunology, Cell Biology, Clinical Laboratory Science, Reagent Systems, Assays, Workflow Development, Instrument Support, GLP

Data Analysis & Visualization – Statistical Analysis, Experimental Data Analysis, Data Interpretation, Power BI, Tableau, R, DAX, Power Query

Programming & Scripting – Python, Java, Spring Boot, SQL, R, Flask, FastAPI, pandas, NumPy, scikit-learn

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Kubernetes

Development & Operations – CI/CD, Jenkins, Git, Agile/Scrum, Jira, Confluence, Splunk, Dynatrace

Enterprise Systems & Integration – REST APIs, ERP Integration Support, Database Integration (PostgreSQL, MySQL, Oracle)

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Automated Assay Performance Monitoring Dashboard

- Developed an **automated dashboard** using **Power BI** to monitor key performance indicators for laboratory assays, visualizing trends and potential issues.

- Integrated data from multiple sources including LIMS and instrument logs via **Power Query** and **DAX** measures, enabling real-time performance tracking.
- Published interactive reports for scientific stakeholders, facilitating quicker identification of assay drift and supporting data-driven decisions for experimental adjustments.

Python-based GLP Data Analysis Pipeline

- Engineered a **Python**-based data pipeline to automate the processing and analysis of experimental results for regulatory submission, ensuring compliance with **GLP** standards.
- Implemented robust error handling and logging mechanisms using **Flask** and **SQLAlchemy**, reducing manual data validation effort by **40%**.
- Designed and built reusable modules for statistical analysis and report generation, enhancing consistency and accuracy across multiple research studies.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)